

Q.3 - Find an equation of the subspace  $W$  of  $\mathbb{R}^3$  spanned by the set of vectors

(i).  $\{(1, -3, 5), (-2, 6, -10)\}$

(ii).  $\{(1, -3, 2), (-2, 0, 3)\}$

(iii).  $\{(1, -2, 1), (-2, 0, 3), (3, -2, -2)\}$

Solution: - (i). The vector  $(x, y, z) \in \mathbb{R}^3$  is in the span of the set  $\{(1, -3, 5), (-2, 6, -10)\}$  if for  $a, b \in \mathbb{R}$

$$(x, y, z) = a(1, -3, 5) + b(-2, 6, -10)$$

$$(x, y, z) = (a - 2b, -3a + 6b, 5a - 10b)$$



we get

$$x = a - 2b$$

$$y = -3a + 6b = -3(a - 2b)$$

$$\text{and } z = 5a - 10b = 5(a - 2b)$$

Put  $a - 2b = t$ , we get

$$x = t, \quad y = -3t, \quad z = 5t$$

are the required equations of subspace.

Spanned by the given set.

(ii) The vector  $(x, y, z) \in \mathbb{R}^3$  is in the span of the set  $\{(1, -3, 2), (-2, 0, 3)\}$  if for  $a, b \in \mathbb{R}$

$$(x, y, z) = a(1, -3, 2) + b(-2, 0, 3)$$

$$(x, y, z) = (a - 2b, -3a, 2a + 3b)$$

we get

$$x = a - 2b \rightarrow (1)$$

$$y = -3a \rightarrow (2)$$

$$z = 2a + 3b \rightarrow (3)$$

From (2), we get

$$a = -\frac{y}{3}$$

Put in (1), we get

$$x = -\frac{y}{3} - 2b$$

$$2b = -x - \frac{y}{3}$$

$$b = -\frac{x}{2} - \frac{y}{6}$$

Put  $a$  and  $b$  in (3), we get



$$z = 2\left(-\frac{y}{3}\right) + 3\left(-\frac{x}{2} - \frac{y}{6}\right)$$

$$z = -\frac{2y}{3} - \frac{3x}{2} - \frac{y}{2}$$

Multiplying both sides by 6, we get

$$6z = -4y - 9x - 3y$$

$$6z = -9x - 7y$$

$$9x + 7y + 6z = 0$$

is the required equation representing the subspace spanned by the given set.

(iii). The vector  $(x, y, z) \in \mathbb{R}^3$  is in the span of the set  $\{(1, -2, 1), (-2, 0, 3), (3, -2, -2)\}$  if for  $a, b, c \in \mathbb{R}$

$$(x, y, z) = a(1, -2, 1) + b(-2, 0, 3) + c(3, -2, -2)$$

$$(x, y, z) = (a - 2b + 3c, -2a - 2c, a + 3b - 2c)$$

we get

$$x = a - 2b + 3c \rightarrow (1)$$

$$y = -2a - 2c \rightarrow (2)$$

$$z = a + 3b - 2c \rightarrow (3)$$

By  $2(1) + (2)$ , we get

$$2x + y = \cancel{2a} - 4b + 6c - \cancel{2a} - 2c$$

$$2x + y = -4b + 4c$$

$$\frac{2x + y}{4} = -b + c \rightarrow (4)$$



By ① - ③

$$x - z = \cancel{a} - 2b + 3c - \cancel{a} - 3b + 2c$$

$$x - z = -5b + 5c$$

$$\frac{x - z}{5} = -b + c \rightarrow \textcircled{5}$$

From ④ and ⑤, we get

$$\frac{2x + y}{4} = \frac{x - z}{5}$$

$$10x + 5y = 4x - 4z$$

$$10x - 4x + 5y + 4z = 0$$

$$6x + 5y + 4z = 0$$

is the required equation representing the  
Subspace spanned by the given set.